

On RQ2 — This is the most important one to understand before you rewrite it. Ask him: *"When you say reframe around a lending decision outcome, do you mean something like approval rate or default prediction accuracy, rather than asking whether the feature selection method improves model metrics?"* You want to confirm what he means by "lending decision outcome" specifically.

On the Home Credit / mortgage alignment — The dataset is for consumer loans in Asia and Eastern Europe, not Canadian mortgages. Ask him: *"Should I reframe the beneficiary as a consumer lending risk team rather than a mortgage team, or should I keep the Canadian banking context and just be more careful about distinguishing it from the dataset's actual population?"*

On the EDA report — He asked you to generate one and upload it to GitHub by the end of this week. Ask him what level of depth he expects: basic distributions and missing value counts, or something more structured? Also clarify if he wants the GitHub repo to hold all future milestones.

On the ethics/jurisdiction gap — Ask him if he wants you to explicitly acknowledge that the dataset is from a non-Canadian lender (Home Credit, Eastern Europe/Asia) and discuss what limitations that creates for applying findings to a Canadian regulatory context.

Changes to Make to the Document

There are six main things to fix:

1. Reframe RQ2 Right now RQ2 is essentially asking: *"Does my feature selection method improve model scores?"* He wants it to ask something more meaningful to a bank, like: *"Does a consensus-selected feature set improve the accuracy of identifying high-risk applicants in a way that would change lending decisions?"* The question should connect the method to a real-world outcome — approving/denying a loan, reducing default rate — rather than ending at a model metric.

2. Fix the mortgage/Home Credit mismatch In Section 1.2 you mention mortgage requests, but Home Credit doesn't do mortgages — they do personal consumer loans to unbanked people in Eastern Europe and Asia. You need to either change the lending context to "consumer lending" or "personal loans," or explicitly frame the project as *simulating* how these methods could apply to a Canadian context while being transparent that the data comes from a different population.

3. Add a provenance and license table He wants a formal table in Section 3 that lists, for each data source: the file name or source path, the data owner, why it was originally collected, the license terms, what population it covers, and the date you accessed it. Right now you have this scattered in prose — he wants it structured and explicit.

4. Expand the ethics section Section 3.4 currently focuses on protected attributes and proxy variables, which is good. But he wants you to also address three things you're missing:

- **Representativeness:** The dataset is from Eastern Europe/Asia, not Canada — does it fairly represent the population you're claiming to help?
- **Misuse risk:** What could go wrong if someone used your model irresponsibly or deployed it in the wrong context?
- **Jurisdiction fit:** Canadian law (OSFI, Human Rights Act) doesn't technically apply to Home Credit's data — you need to explicitly acknowledge this gap and explain how you're bridging it for the simulation.

5. Make the GenAI verification plan specific Right now Section 6 basically says "I'll document my AI use." He wants you to describe *how* you'll verify each type of output. For example: how will you verify that code generated with AI help actually works correctly? How will you verify that citations are real and accurate? How will you verify that written claims reflect what the data actually shows? Each category needs its own concrete check.

6. Create and upload an EDA report to GitHub This isn't a document change but it's part of what he's asking for. You need to run some initial exploratory data analysis — distributions of key variables, missing value counts, class imbalance visualization, correlation heatmap — and push it to a GitHub repo, then invite him as a collaborator. He set this as a deadline for end of this week

Here are your 5 questions, ranked by priority:

1. **RQ2** — "When you say reframe around a lending decision outcome, do you mean something like approval rate or default prediction rate, rather than asking whether the feature selection method improves model scores?"
2. **Home Credit / mortgage mismatch** — "Should I reframe the beneficiary away from mortgage banking to consumer lending, or keep the Canadian context and just acknowledge the dataset is from a different population?"
3. **Ethics/jurisdiction gap** — "For the jurisdiction section, do you want me to explicitly acknowledge that OSFI and the Human Rights Act don't apply to Home Credit's data and explain how I'm bridging that gap for the simulation?"

4. **EDA report** — "What depth are you expecting for the EDA — basic distributions and missing values, or something more structured? And do you want all future milestones in the same GitHub repo?"
5. **GenAI verification** — "For the verification plan, do you want a specific check described for each type of AI use, like code, citations, and written claims separately?"

What the prof mentioned during the meeting:

- to remove the solution from the question and also not make the questions datadriven but to focus on a business problem and then use the data to analyze the problem but the analysis has to be done by me
- remove buzzwords or explain them in a glossary (like 'default', other bank terms)
- the research questions need to be written focused on the problem not including the solution.
- who will benefit? what department? what will they do with the solution
- what is the main problem?
- He wanted me to look at the population for the data to see where they are from. also what decade the data is from what year? is it still relevant?
- what are the boundaries and limitations to the dataset
- He also wants specificity in what do i mean by 'stronger'?
- he wants me to make solid assumptions like a 80% threshold, or whatever the bank industry standards are
- The pipeline needs to be specific
- in the analysis: he wants the subpopulations- demographic factors and time frames
- Investigate stratified clustering and then compare
- Time frame for the data we are looking at
- He wants me to be granular, when i mention risk of default he wants me to look not just for risky or not risky but to go deeper at types of risk or percentage of risk.
- He also wants me looking at a bigger confusion matrix saying it could be even a 5by5 or bigger.
- for the eda report which is what i am working on now for monday he wants me to look at wideit and sweep visualization libraries and for the EDA report to be in HTML
- in the EDA report he wants multivariate, univariate analysis, types of data for each feature, skewness, correlation, missing ratio, uniformity.
- I also need to use a Journal that includes the question i am solving for. the approach. the tools i used to solve
- I also need a second journal for AI use: he wants me to be more specific about how am i verifying answers. how am i using it. whats the plan.

- in Ethics he wants me to keep track of permissions and tools to make sure they are okay to use.
- He wants 3 compelling questions
- finally he wants my literature review:such as= 'replacing missing values'; which method should i use with the reasoning behind it. did i use mean, median,mode, etc...